

13. Explain the concept of green computing. How can computing professionals contribute to environmental sustainability?

Concept of Green Computing

Green computing refers to the environmentally responsive design, use and disposal of computers and related sources. Its goal is to reduce the environmental impact of computing system by minimizing energy consumption, reducing electronic waste and promoting sustainable practices.

Objectives:

- Reducing energy consumption
- Lower carbon emission,
- Minimize e-waste.
- promote recycling and reuse.

Ways computing professionals can contribute

- Develop software and hardware that use less power, optimize processing and reducing unnecessary resource consumption.
- Use efficient cooling systems and cloud optimization to reduce electricity use.
- Promote digital communication, online documentation and remote collaboration to reduce paper use.
- Write optimized code that reduces processing time, memory usage and energy consumption.
- Supports systems powered by renewable energy sources.

14. Explain Virtue Ethics and describe how it can guide ethical decision making for software engineers.

Virtue ethics and its role in software engineering:

Virtue ethics is an ethical framework that focuses on the character and moral virtues of a person rather than rules. It emphasizes developing good qualities such as: honesty, responsibility-fairness and empathy. The idea originates from Aristotle, who argued that ethical behaviour come from being a good person.

Guidance for software engineer:

Virtue ethics guides software engineers to ask what would a good and responsible engineer do?

Instead of only following rules, engineers are encouraged to:

1. Write reliable and secure code.
2. Be honest about software limitations and risks.
3. Respect user privacy and data.
4. Take responsibility for the social impact of the software.

15. What are the key principles of the ACM code of ethics? Why are such professional codes important for computing professionals?

The ACM code of ethics provides ethical guidelines for computing professionals. Its key principles include -

- Contribute to society and human well being by ensuring technology benefits people.
- Avoid harm such as: privacy violations, security risks or unsafe systems.
- Respect privacy and confidentiality of users data.
- Maintain professional competence and follow quality standards.
- Accept responsibility and accountability for one's work.

Importance of professional codes for computing professionals

- Provide clear ethical guidance for decision making in complex situations.
- Promote public trust in computing technologies.
- Help professionals balance duties to employees, users and society.
- Encourage responsible, fair and lawful conduct in software development and use.

16. The ACM code of ethics emphasizes both contribute to society and human well-being and respect privacy. Describe a realistic scenario where these two principle might come into conflict for a data scientist.

Conflict between "contribute to society" and "respect privacy".

Scenario:

A data scientist working for a public health organization analyzes large-scale mobile location and health data to predict the spread of an infectious disease. Using detailed personal data could significantly help society by improving early warnings and saving lives. However, this data contains sensitive personal information and analyzing it may violate individuals' privacy. Here, the principle of contributing to society

and human well-being conflicts with the principle of respecting privacy!

How the ACM code of ethics guides resolution

The ACM code of ethics guides the professionals to -

- Minimize harm by using the least intrusive data necessary.
- Apply anonymization and aggregation to protect individual identities.
- Ensure transparency and informed consent where possible.
- Use data only for the stated public benefit and prevent misuse.

17.
23. What is a conflict of interest in the IT workplace? Provide Examples and explain how professionals should ethically handle such situations!

Conflict of Interest in the IT workplace

A conflict of interest occurs when a professional's personal interests, financial benefits or relationships interfere with their professional duties and responsibilities.

In the IT workplace, this happens when an employee's private interest could influence their objective judgment or decision-making.

Examples of conflict of interest in IT:

1. **Financial interest:** An IT manager recommends purchasing software from a company in which they own shares without sharing the relationship.
2. **Personal relationships:** A project leader hires a

friend or relative without fair evaluation of other candidates.

3. Dual employment: When a software developer works for two competing companies at the same time.

4. Accepting Gifts: A vendor offers expensive gifts to an IT officer to secure a contract.

5. Misuse of sensitive info: An employee uses sensitive info of a company to benefit another business.

How professionals should ethically handle conflict of interest:

1. Immediately inform the management about any potential conflict.

2. Be open about financial interests that may affect decisions.

3. Step away from decision making where partiality is not compromised.

4. Make decisions based on fairness and professional

responsibility, not personal gain.

18. Explain the concept of justice and fairness according to any one religions traditional ideology. How can these principles be applied to promote ethical decision making in modern professional life?

Concept of Justice and fairness in Islam

In Islamic tradition, justice is a foundational concept and divine attribute of Allah. It is defined as placing something in its rightful place and ensuring that every individual receives their right without discrimination.

- The Quran explicitly commands believers to stand out firmly for justice, as witness to Allah, even as against anyone.
- Islamic justice is universal and applies to all people regardless of race, status or creed.
- Justice is seen as equilibrium, it requires giving everyone their 'rightful due' without

excess, delay or misery.

• Islamic justice always focus on your intentional intentions; as individuals are always in the presence of god.

Application in Modern Professional Life

- Avoid discrimination in hiring or promotion.
- Ensure equal pay for equal work.
- Avoid fraud, bribery and corruption.
- Provide accurate information to clients and customers.
- Maintain transparency in financial and professional dealings.
- Avoid actions that can cause harm to individuals or communities.

19. What is Aristotle's concept of the Golden Mean? and how does it guide individuals to find a balanced and virtuous course of action between extremes in ethical decision making?

Aristotle's concept of the golden Mean

The Golden Mean is a moral principle proposed by Aristotle. It states that virtue lies between two extremes: one of excess and one of deficiency. According to Aristotle, ethical behaviour is not about rigid rules but about finding the right balance in a particular situation using reason. In simple terms -

- Too much of something = vice (excess)
- Too little of something = vice (deficiency)
- The right balance = virtue (Golden Mean)

How it guides ethical decision making

1. Encourages Balance:

Individuals are guided to avoid extreme actions and instead seek moderation.

2. Uses practical wisdom:

Aristotle emphasized the importance of rational judgement to determine what is appropriate in each situation.

3. Considers context

The "Mean" is not the same for everyone it depends on individual roles.

4. Builds moral character

By consistently choosing balanced actions, individuals develop virtuous habits.

20. Discuss the issue of moral decay in BD by analysing its major causes, social and economic effects and possible strategies to address & prevent it.

Moral decay refers to the deterioration of ethical values, social norms and moral behaviour in a society. In BD, many believe moral decay threatens social harmony, trust and national development.

Major causes

1. Weak enforcement of law and order.
2. People see bribery and nepotism as normal, weakening moral standards.
3. Breaking down Family and Community values.
4. Social media spreads misinformation and encourages impulsive behaviour.
5. Economic pressure and Inequality.
6. Lack of ethical education.

Social and economic effects of moral decay

- Breakdown of trust in communities.
- Erosion of respect for elders and societal norms.
- Reduced foreign investment due to poor governance and corruption.
- Increased cost of doing business because of bribery and inefficiency.
- Brain drain due to systemic moral failure.

Strategies to address and prevent it

1. Strengthen law enforcement and anti-corruption measures.
2. Promote moral and civic education.
3. Encourage ethical leadership.
4. Increase employment opportunities and social awareness.

21. Examine how Animal Farm reflects the dangers of unchecked power and moral responsibilities of leaders.

Animal Farm is a political allegory/metaphor that illustrates how unchecked power leads to corruption and betrayal of ideals. The story portrays farm animals who overthrew their human owner to create a society based on equality, but it ultimately results in a dictatorship.

Dangers of unchecked power

1. Corruption of leadership: After the rebellion, the pigs - especially Napoleon gradually take control. Without accountability, power corrupts them. They change the rule that "All animals are equal" to "All animals are equal, but some animals are more equal than others."

2. Suppression of opposition: Napoleon eliminates rivals like Snowball and uses fear, propaganda and violence to silence opposition.

3. Manipulation and propaganda: They spread misinformation to control the animals. The pigs change history and rewrite laws to justify their action. This shows how political manipulation can deceive people when they lack awareness.

Betrayal of equality

- The pigs enjoy privileges.
- The working animals suffer more than before.
- The leadership become the same like the humans.

Though the revolution began with the goal of equality but it led to betrayal of equality.

Moral responsibilities of leaders

The novel emphasizes that leaders have moral duties to-

- Serve the common good
- Remain accountable
- Protect equality and justice
- Avoid abuse of power.

Napoleon fails in all these responsibilities, showing that leadership without morality destroys both freedom and unity.

22. Discuss the role of mentorship and personal growth in shaping Simba's ability to confront challenges and assume leadership.

In the Lion King, Simba's journey from a carefree cub to a responsible king highlights the importance of mentorship and personal growth. His transformation shows that leadership is not inherited automatically; it is developed through guidance, experience and self-discovery.

Role of early mentorship

- Mufasa - teaching responsibility: Mufasa teaches Simba about courage, responsibility and the "circle of life". He shows that a true king protects his kingdom and thinks of others first. These lessons later guide Simba in difficult times.
- Rafiki - encouraging self-realization: Rafiki helps Simba understand that he can't escape his past. He reminds Simba that learning from mistakes is necessary.

for growth. This pushes Simba to accept his identity and destiny.

• Timon and Pumbaa - Emotional support:

Timon and Pumbaa provide friendship and comfort. Although their "Hakuna Matata" lifestyle helps Simba heal, it also delays his acceptance of responsibility. This contrast highlights the importance of balanced guidance.

Personal growth:

1. Facing guilt: Simba overcomes the guilt of his father's death.
2. Accepting identity: He realizes he is a rightful king.
3. Building courage: He returns to Pride Rock to confront Scar.

And finally he chooses to protect his kingdom.

Leadership development:

Simba's mentors shape his values, while personal experience strengthens his character. He understands that leadership requires -

• Courage

- Accountability
- Compassion
- Sacrifice

23. How do characters like Mr. Burns and Homer in Simpsons illustrate the ethical consequence of greed and selfishness and what lessons can computing professionals draw?

The Simpsons uses satire to highlight ethical issues in society. Characters like Mr. Burns and Homer Simpson illustrate how greed and selfishness can negatively affect individuals and communities.

Mr. Burns: The extreme of Corporate Greed

Mr. Burns owner of the Springfield Nuclear Power Plant, represents unchecked corporate power and profit driven decision making.

Ethical consequence of his greed

- Exploitation of workers - prioritizes profit over employee welfare.
- Environmental damage - neglects safety regulations, risking public health.

- Lack of accountability - avoids responsibility for harmful actions.

Lesson for computing professionals

In IT organizations, prioritizing profit over safety can lead to:

- Ignoring security vulnerabilities
- Releasing unsafe software
- Misusing user data

Homer Simpson - selfishness and carelessness

Homer works on nuclear plant but often lazy and focused on personal comfort.

Ethical consequence

- Carelessness in a critical job.
- Poor decision making.
- Risking public safety.

Lesson for computing Professionals

- A careless programmer may introduce security flaws.
- Ignoring testing may cause system failures.

- weak password practices can compromise entire networks.

24. What is Digital Divide? Discuss its social implications and suggest measures to bridge the gap.

Digital Divide:

The digital divide refers to the gap between individuals, communities or countries that have access to modern digital technologies and those that do not. This divide may exist due to differences in income, education, location, gender.

Social Implications

- Students without internet access can't derive benefit from online learning resources, creating unequal academic opportunities.
- People lacking digital skills to find jobs, use online banking in digital markets.
- Without digital access, individuals miss out on important news, govt services and health info.

The divide increase inequality between urban and rural populations and between developed and developing countries.

Measures to bridge the gap:

1. Expand affordable internet infrastructure
2. Provide low cost devices for students and low income families
3. Promote digital literacy programs in schools and communities.
4. Govt and private sector partnership.
5. Encourage inclusive technology policies.

25. Discuss the role of ethical leadership in IT organizations. How can leaders foster an ethical culture within their teams?

Role of Ethical Leadership

1. Ethical leaders build trust among employees, clients and stakeholders by being transparent and honest.
2. They ensure that the organization follows laws, data protection policies and professional codes of conduct.
3. In IT, leaders must prioritize cybersecurity, data privacy and responsible use of technology.
4. Ethical leaders ensure that new technologies don't harm society.
5. Strong ethical guidance prevents fraud, plagiarism, data misuse and unfair practice.

How it guides ethical decision making

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